

Friday 5/8 Lectures: Bone Marrow and Hematologic Malignancies

- Block: Hematology
- Method: Codex-authored synthesis from extracted lecture markdown in this workspace.
- Sources:
 - 20210309_Borowitz_Normal marrow electure.pdf
 - 20210309_Brodsky_Bone_Marrow_Failure_Syndromes.pdf
 - 2026_Yui_Introduction_to_Heme_Malignancies.pdf
 - 2026_Yui_Laboratory_Methodologies_for_Heme_Malignancies.pdf
 - Yui_Hematologic_Malignancies_Flowchart.pdf
 - 20210311_Borowitz_Acute_Leukemias.pdf
 - 2025_Imus_Acute_Lymphoblastic_Leukemia.pdf
 - 20220314_Borowitz_Myelodysplastic_Syndrome.pdf
 - 2025_Yui_Myeloproliferative_Neoplasms.pdf
 - 2025_Yui_Chronic_Myeloid_Leukemia.pdf

One-Minute Mental Model

This lecture set is the bridge from **normal marrow** to **marrow failure** to **clonal hematologic malignancy**.

Start every case with three questions:

1. **Is the marrow producing normally?** Look for cytopenias, blasts, dysplasia, fibrosis, and whether cellularity is appropriate for age.
2. **Which lineage is abnormal?** Myeloid, lymphoid, erythroid, megakaryocytic, or multilineage.
3. **Is the process acute, chronic, proliferative, dysplastic, or aplastic?**

The big organizing rule:

- **Acute leukemia** = blasts, rapid marrow failure, urgent.
- **Chronic leukemia/MPN** = more mature cells, proliferation, often high counts.
- **MDS** = ineffective dysplastic hematopoiesis, cytopenias, AML risk.
- **Aplastic anemia/bone marrow failure** = hypocellular marrow and pancytopenia, often immune or inherited stem-cell vulnerability.
- **Lab methods** define the clone: morphology, flow cytometry, karyotype/FISH, and NGS each answer different questions.

High-Yield Visual Map

Normal Bone Marrow

Normal marrow is best understood as an organized factory. Core biopsy shows **cellularity and architecture**; aspirate smear shows **cell detail and maturation**.

What To Recognize

- Marrow biopsy is best for overall cellularity, architecture, fibrosis, and focal infiltrates.
- Aspirate smear is best for cell morphology, maturation sequence, blasts, and dysplasia.
- Normal adult marrow has a mixture of fat and hematopoietic cells.
- Myelopoiesis begins near bony trabeculae; erythroid islands tend to form around central macrophages.
- Megakaryocytes are large marrow cells with multilobated nuclei and platelet-producing cytoplasm.
- Plasma cells and lymphocytes are normally present only as small background populations.

Rapid Normal Marrow Checks

Feature	Normal idea	Why it matters
Cellularity	Roughly age-adjusted cellularity with fat	Hypocellular suggests aplastic/failure; hypercellular suggests proliferative or infiltrative process
Myeloid maturation	Full spectrum from blasts to neutrophils	Maturation arrest or excess blasts suggests leukemia/MDS

Feature	Normal idea	Why it matters
Erythroid maturation	Nucleated precursors maturing toward reticulocytes	Dysplasia, megaloblastosis, or erythroid predominance narrows differential
Megakaryocytes	Scattered large cells	Small hypolobated forms suggest MDS; clustering/atypia can suggest MPN
Blasts	Low background population	Increased blasts drive acute leukemia/MDS classification

Bone Marrow Failure Syndromes

Bone marrow failure means inadequate hematopoiesis, usually with cytopenias. The key split is **congenital/inherited** vs **acquired**.

Congenital Bone Marrow Failure

Syndrome	Core defect	High-yield clues
Fanconi anemia	DNA repair defect	Congenital anomalies, marrow failure, AML/MDS risk
Dyskeratosis congenita	Telomere maintenance defect	Nail dystrophy, abnormal skin pigmentation, oral leukoplakia, pulmonary/liver risk
Diamond-Blackfan anemia	Ribosomal disorder affecting erythropoiesis	Pure red-cell aplasia, macrocytic anemia, congenital anomalies

Acquired Aplastic Anemia

- Usually immune-mediated destruction/suppression of hematopoietic stem cells.
- Presents with pancytopenia and hypocellular marrow.
- It is not a primary platelet, RBC, or neutrophil problem; it is a stem-cell production problem.
- Severe aplastic anemia is treated with hematopoietic stem cell transplant when appropriate or immunosuppression, often antithymocyte globulin plus cyclosporine.
- PNH, aplastic anemia, MDS, and AML are biologically connected because immune selection and stem-cell mutations can reshape the hematopoietic clone.

Introduction To Hematologic Malignancies

Hematologic malignancies are clonal diseases of blood, marrow, lymph node, spleen, or plasma cells. The name often tells you **lineage** and **clinical tempo**, but not perfectly.

Core Classification Logic

Axis	Question	Examples
Lineage	Myeloid or lymphoid?	AML/CML/MDS/MPN vs ALL/CLL/lymphoma
Maturity	Blast/immature or mature?	Acute leukemia vs chronic leukemia/lymphoma
Site	Blood/marrow or tissue mass?	Leukemia vs lymphoma, with overlap
Biology	Proliferation, dysplasia, survival, plasma cell clone?	MPN, MDS, low-grade lymphoma, myeloma

Important caveat: leukemia vs lymphoma is partly a presentation pattern. A lymphoid malignancy can have both blood/marrow and nodal/tissue phases.

Acute vs Chronic Leukemia

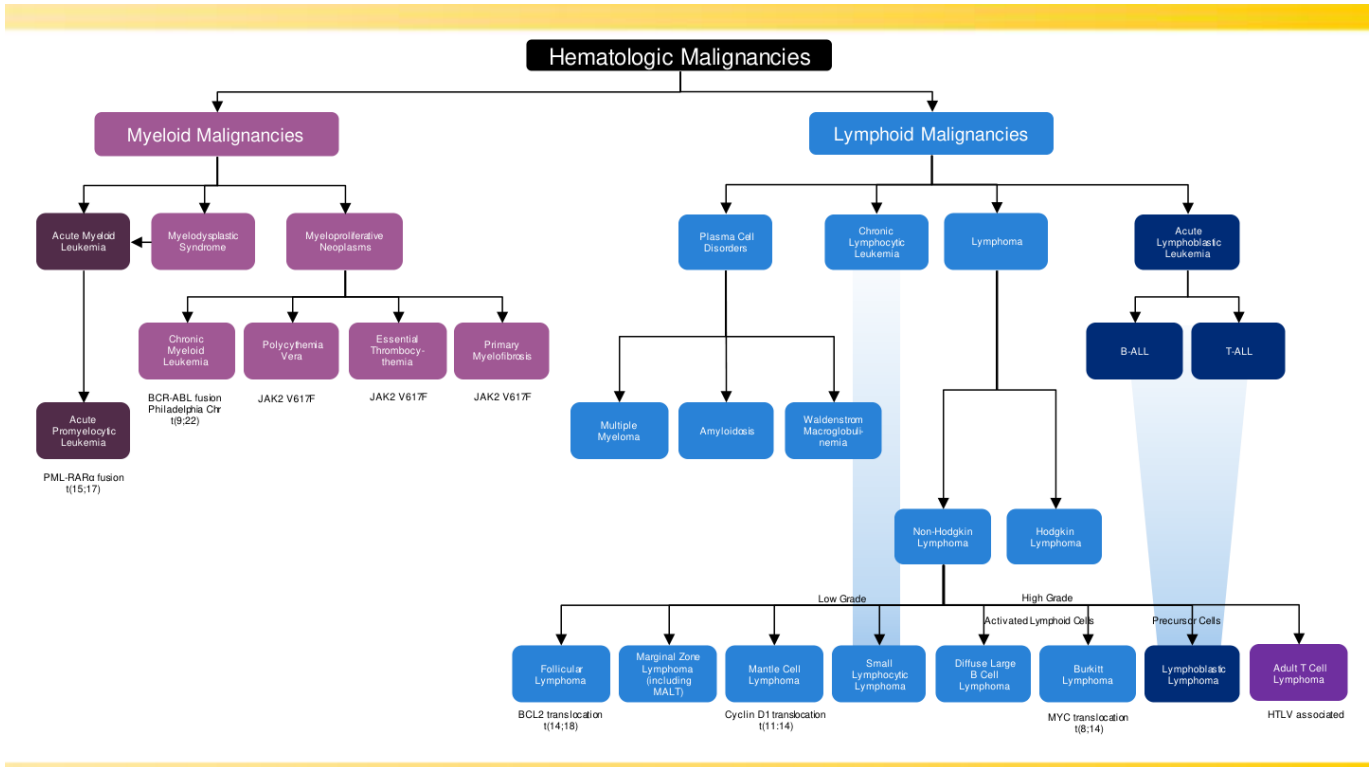


Figure 1: Final hematologic malignancies flowchart. Use this as the master classification map for myeloid vs lymphoid malignancies, acute vs chronic processes, and major disease families.

Normal Hematopoiesis

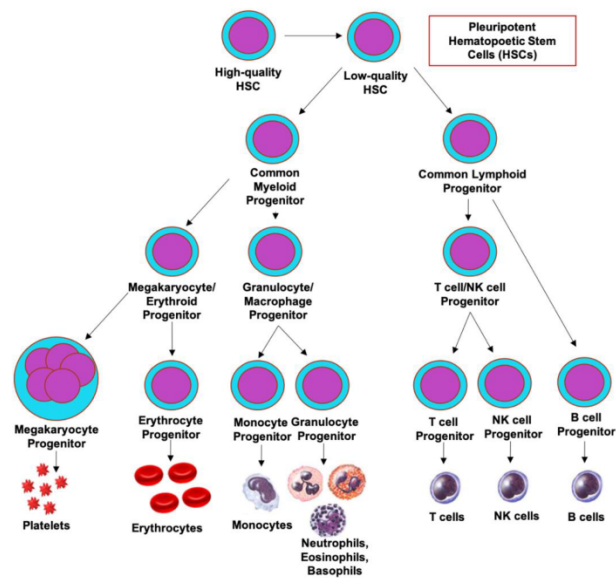
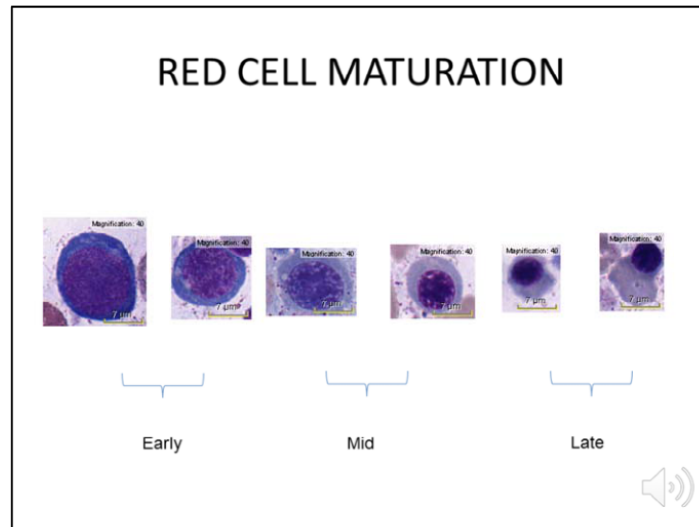
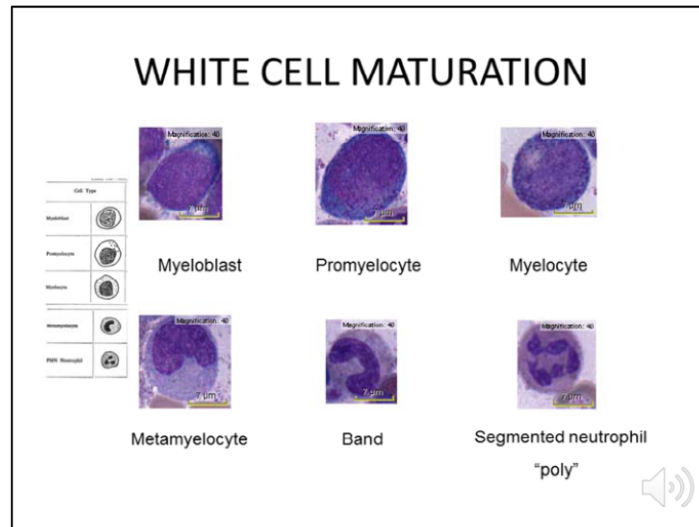


Figure 2: Normal hematopoiesis overview.



Maturation is easier to see in the Wright-stained preparations. From the illustration you can see that there is a continuous process in which the cells get smaller, the cytoplasm goes from dark blue to blue gray, to gray and sometimes to pink. There are also important nuclear changes with the earliest ("blast") having smooth chromatin (euchromatin). As they mature, the chromatin condenses in a characteristic chunky pattern until eventually it is a homogeneous dark circle, at which point it gets extruded from the cells. Conventionally we divide red cell maturation into 3 stages; terminology is not important and a couple of different nomenclatures used: Early, mid and late will suffice.

Figure 3: Erythroid maturation in normal marrow.



White cell maturation is also easier to appreciate on Wright stained material. The myeloblast has very smooth chromatin, bluish cytoplasm and no granules, the latter appearing in the promyelocyte stage. Myelocytes have a Golgi but still have granules and a round to oval nucleus, while the nucleus indents significantly at the metamyelocyte stage. As cells mature cytoplasm goes from blue to blue-gray, to gray to pink. As you already have learned, the segmented neutrophil is distinguished from the band by the pinching off of nuclear lobes. Note the diagram of myeloid maturation taken from your handout. That diagram combines the metamyelocyte and band stage but these are generally separated, with bands having smaller nuclei and more condensed (darker) chromatin. As with red cell maturation, there are continuous changes that we arbitrarily divide into stages.

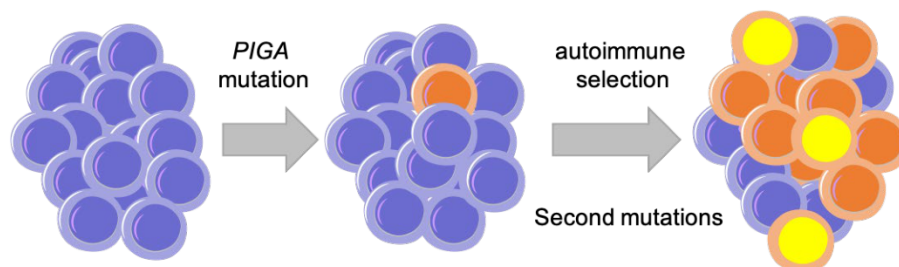
Figure 4: Myeloid maturation in normal marrow.

Bone Marrow Failure Syndromes

- Congenital (inherited) syndromes
 - Fanconi Anemia (FA)
 - Dyskeratosis Congenita (DC)
 - Diamond-Blackfan Anemia (DBA)
- Acquired Aplastic Anemia syndromes
 - Medication-related
 - Infection-related
 - Idiopathic (Immune-Mediated)

Figure 5: Bone marrow failure syndrome categories.

Relationship between Aplastic Anemia, PNH, and Acute Leukemia



PIGA mutations protect hematopoietic stem cells from immune mediated destruction in aplastic anemia

Autoimmune attack also predisposes to acquisition of additional mutations that can lead to leukemia (AML/MDS)

Figure 6: Relationship between aplastic anemia, PNH, and acute leukemia.

The Four Leukemias (a Misnomer)

	Myeloid	Lymphoid
Acute	Acute Myeloid Leukemia	Acute Lymphoblastic Leukemia
Chronic	Chronic Myeloid Leukemia	Chronic Lymphocytic Leukemia

Chronic myeloid leukemia (CML) is a myeloproliferative neoplasm (MPN).
Chronic lymphocytic leukemia (CLL) is the leukemic phase of a low grade lymphoma.

Figure 7: The four leukemias are useful but oversimplified.

Category	Cell maturity	Tempo	Classic pattern
Acute leukemia	Immature blasts	Days to weeks	Marrow failure, infections, bleeding, fatigue, bone pain
Chronic leukemia	More mature cells	Months to years	High counts, splenomegaly, sometimes incidental

Laboratory Methodologies

No single lab method gives the whole answer. Each method views the malignancy from a different angle.

Take Home Points- Summary Table

	Flow Cytometry	Karyotype	FISH	NGS
What is measured?	Protein expression (surface or intracellular) Immunophenotype	Whole chromosomes (structure and number) Cytogenetics	Specific DNA sequences Cytogenetics	DNA sequence/ gene mutations
Methodology	Fluorescent Ab labelling, laser-based cell analysis	Chromosome banding and visualization	Fluorescent DNA probes hybridized to target sequences	Massively parallel DNA sequencing
Cell requirements	Viable cells	Dividing cells (metaphase)	Dividing or non-dividing cells	DNA (does not require viable cells)
Speed	Hours	Days	1-2 Days	Days/weeks
Clinical Role	Lineage, maturation, clonality	Disease ID, prognosis	More rapid for specific lesions than karyotype	Prognosis, targeted therapy

Figure 8: Summary table of flow cytometry, karyotype, FISH, and NGS.

Acute Leukemias

Acute leukemia presents when blasts overtake marrow function. Symptoms often come from marrow failure: anemia, neutropenia, thrombocytopenia, and bone pain. Organ infiltration can add lymphadenopathy, splenomegaly, gingival hypertrophy, skin lesions, or CNS/testicular disease depending on subtype.

Distinguishing ALL From AML



	ALL	AML
Morphology	Nothing specific	Granules; Auer rods
Enzymology	TdT	Myeloperoxidase
Flow Cytometry	B: CD19, CD10, cCD79a T: CD7, CD5, cCD3	CD33, CD13, CD117

Figure 9: Distinguishing ALL from AML using morphology, cytochemistry, and flow cytometry.

AML vs ALL

Feature	AML	ALL
Lineage	Myeloid blasts	Lymphoblasts, usually B or T
Morphology	Granules or Auer rods can appear	Often no specific morphology
Cytochemistry	Myeloperoxidase positive	TdT positive
Flow examples	CD13, CD33, CD117	B: CD19, CD10, CD79a; T: CD7, CD5, cytoplasmic CD3
Clinical emphasis	Adults, APL/DIC emergency, AML genetics	Children more common, CNS sanctuary sites, Ph+ subset

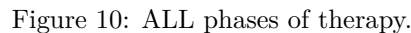
AML Anchors

- AML is a clonal myeloid blast disorder with impaired differentiation.
- Acute promyelocytic leukemia is a special emergency because it can cause severe DIC; treat promptly with ATRA-based therapy.
- AML genetics matter for prognosis and therapy. Recurrent lesions and cooperative mutations drive classification.
- AML can arise de novo or evolve from MDS or prior therapy.

ALL Anchors

- ALL is a lymphoblast malignancy; B-ALL is most common, T-ALL often presents with mediastinal mass.
- Childhood ALL has excellent cure rates; adult ALL is higher risk.
- CNS and testes are sanctuary sites, so CNS-directed prophylaxis is part of therapy.
- Ph+ ALL has BCR-ABL and is treated with a tyrosine kinase inhibitor plus chemotherapy.

Phases of Therapy



- Presentation: fatigue, fever, infection, bruising/bleeding, bone pain, lymphadenopathy, hepatosplenomegaly, or mediastinal mass.
- Diagnosis: marrow/peripheral blasts plus flow cytometry to define B vs T lineage.
- Workup includes cytogenetics/molecular testing, CNS evaluation, and risk stratification.
- Risk features include age, WBC count, cytogenetics, Ph status, response to induction, and measurable residual disease.

Phase	Purpose
Induction	Achieve remission
Consolidation/intensification	Eradicate residual disease
Interim maintenance	Continue pressure while managing toxicity
Delayed intensification	Re-intensify therapy
Maintenance	Long-term suppression, often 2-2.5 years

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Myelodysplastic Syndrome

MDS is ineffective, dysplastic hematopoiesis. The marrow can be hypercellular, but production is poor, so patients have cytopenias.

Diagnosis of MDS: Summary

- Morphology is the cornerstone of diagnosis
- Prognostic scores combine morphology with peripheral blood findings and cytogenetics
- Molecular profiling adds prognostic information

Figure 11: Diagnosis of MDS summary.

What Defines MDS

- Usually older adults with unexplained cytopenias.
- Dysplasia can involve erythroid, myeloid, and/or megakaryocytic lineages.
- Morphology is central, but cytogenetics and molecular profiling add prognosis.
- MDS can progress to AML as additional mutations accumulate.
- Blasts matter: more blasts means higher risk and closer relationship to AML.

MDS High-Yield Table

Concept	Anchor
Clinical clue	Cytopenias despite often cellular marrow
Morphology	Dysplasia, abnormal maturation, possible ring sideroblasts
Prognosis	Blasts, cytogenetics, cytopenia depth, mutations
AML relationship	MDS can transform to AML
Treatment range	Supportive care, growth factors/transfusions, lenalidomide in selected del(5q), hypomethylating agents, transplant for eligible high-risk patients

Myeloproliferative Neoplasms

MPNs are clonal stem-cell disorders with too much mature myeloid production. They are proliferative rather than primarily dysplastic.

Classic BCR-ABL-Negative MPNs

Take Home Points

- The primary pathogenesis of MPNs features mutations (most frequently JAK2 V617F) that lead to constitutive activation of the JAK/STAT pathway and proliferation of mature myeloid elements
- PV, ET, and PMF have different clinical phenotypes, but generally all confer risk of vasomotor symptoms, thrombosis, leukemic transformation
- Therapies including therapeutic phlebotomy and cytoreductive therapies including hydroxyurea, ruxolitinib, and interferon can improve symptoms and risk of thrombosis, and be potentially disease-modifying

Figure 12: MPN take-home points.

Disease	Dominant count	Driver clues	Clinical clues
Polycythemia vera	High RBC mass	JAK2, low EPO	Thrombosis, aquagenic pruritus, erythromelalgia, splenomegaly
Essential thrombocythemia	High platelets	JAK2/CALR/MPL	Thrombosis or bleeding, vasomotor symptoms
Primary myelofibrosis	Fibrotic marrow, extramedullary hematopoiesis	JAK2/CALR/MPL	Splenomegaly, teardrop cells, leukoerythroblastosis, constitutional symptoms

MPN Management Anchors

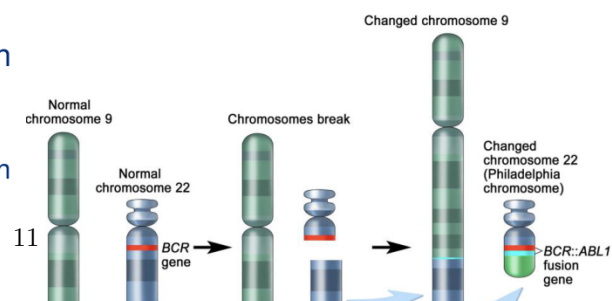
- Main risk in PV and ET is thrombosis.
- Low-dose aspirin is common if no contraindication.
- PV often uses phlebotomy to control hematocrit.
- Cytoreduction options include hydroxyurea, interferon, anagrelide in selected settings, and ruxolitinib for JAK-STAT pathway inhibition.
- MPNs can progress to myelofibrosis or acute leukemia.

Chronic Myeloid Leukemia

CML is the BCR-ABL-driven myeloproliferative neoplasm. It is the cleanest example of a molecular lesion defining a disease and guiding targeted therapy.

Pathophysiology of CML

- Translocation of the ABL1 gene on chromosome 9 with the BCR gene on chromosome 22- t(9;22)
 - Forms the new chromosome 22 (aka **Philadelphia chromosome- Ph**) with **BCR:ABL1 fusion gene**
- The BCR-ABL1 fusion gene leads to



Take Home Points

- CML is a specific MPN caused by a specific translocation (BCR-ABL) in a stem cell which produces a constitutively active tyrosine kinase
- CML presents with leukocytosis with granulocytes at all stages of maturation, often with mild anemia, but preserved or elevated platelet count
- Without treatment, CML will evolve into an accelerated and then blast phase which behaves as acute leukemia- either myeloid or lymphoid
- Imatinib is a tyrosine kinase inhibitor targeting BCR-ABL; it has changed CML from a fatal disease into a chronic, manageable condition

Figure 14: CML take-home points.

Must-Know Comparisons

Problem	Marrow	Blood counts	Key clue	Acute leukemia risk
Aplastic anemia	Hypocellular	Pancytopenia	Immune/inherited stem-cell failure	Can overlap with clonal evolution
MDS	Often cellular, dysplastic	Cytopenias	Ineffective hematopoiesis, dysplasia	Yes
MPN	Hypercellular/proliferative	High mature myeloid counts	JAK-STAT/BCR-ABL driver logic	Yes, variable
AML	Blasts	Cytopenias or high WBC with blasts	Myeloid markers, Auer rods/MPO	Already acute
ALL	Lymphoblasts	Cytopenias or high WBC with blasts	TdT, B/T lymphoid markers	Already acute
CML	Granulocytic proliferation	High WBC with left shift	BCR-ABL, basophilia	Blast crisis

Common Confusions To Avoid

- **Hypercellular marrow does not mean effective production.** MDS can be hypercellular while peripheral counts are low.
- **Aplastic anemia is hypocellular.** The marrow is empty/fatty, not packed with dysplastic or malignant cells.
- **Acute vs chronic is about maturity and tempo, not simply WBC count.**
- **CML is an MPN.** It is separated from PV/ET/PMF because it is BCR-ABL positive.
- **MDS and AML are a continuum.** Rising blasts and clonal evolution move MDS toward AML.
- **Flow cytometry tells lineage.** Cytogenetics/FISH/NGS tell recurrent genetic lesions and prognosis.
- **The flowchart is a map, not a memorization trap.** Use it to orient the disease family before memorizing details.

Rapid Review

- Normal marrow: biopsy for architecture/cellularity; aspirate for morphology.
- Bone marrow failure: inherited or acquired stem-cell production failure; aplastic anemia is hypocellular pancytopenia.

- Aplastic anemia, PNH, MDS, and AML can be connected by immune selection and clonal evolution.
- Acute leukemia: blasts, marrow failure, urgent workup with morphology, flow, cytogenetics, and molecular tests.
- AML: myeloid blasts; APL is DIC emergency; genetics drive prognosis.
- ALL: lymphoblasts, TdT, CNS prophylaxis, long phase-based therapy.
- MDS: ineffective dysplastic hematopoiesis with cytopenias and AML risk.
- MPN: overproduction of mature myeloid cells, thrombotic risk, JAK-STAT pathway logic.
- CML: BCR-ABL, granulocytic leukocytosis with basophilia, TKI-responsive.

Active Recall

1. What does a marrow biopsy show better than an aspirate smear?
2. What distinguishes aplastic anemia from MDS?
3. Why can MDS cause cytopenias despite a cellular marrow?
4. What tests separate AML from ALL?
5. What does flow cytometry answer that karyotyping does not?
6. What does FISH answer that karyotyping may miss?
7. Why is APL an emergency?
8. What are the phases of ALL therapy?
9. Which MPN is BCR-ABL positive?
10. What lab pattern makes CML different from a leukemoid reaction?